



A PROJECT ON
**“Water Pollution & Disease Exploratory Analysis –
Typhoid Cases Dashboard”**

Data Processing & Business Analysis
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UNDER THE GUIDANCE OF

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“Water Pollution & Disease Exploratory Analysis – Typhoid Cases Dashboard”

1. Abstract

Water pollution is one of the major environmental issues affecting public health worldwide. Contaminated water sources can lead to the spread of waterborne diseases such as typhoid. This project aims to analyze the relationship between water quality indicators, water accessibility, environmental factors, and typhoid disease cases using Power BI. The dashboard was developed using the "Water Pollution and Disease" dataset obtained from Kaggle. Various visualizations such as KPI cards, donut charts, treemap, line charts, and bar charts were used to identify trends, patterns, and correlations between water pollution parameters and typhoid cases. The dashboard helps users understand how factors like rainfall, water source type, healthcare access, GDP per capita, urbanization rate, and water treatment methods influence disease prevalence.

2. Keywords

Water Pollution, Typhoid Disease, Public Health, Power BI, Data Visualization, Water Quality, Healthcare Access, Clean Water Access, Environmental Analysis, Exploratory Data Analysis (EDA) .

3. Objectives

- To analyze the impact of water pollution on typhoid disease cases.
- To study water quality indicators such as bacteria count, contaminant level, dissolved oxygen, and lead concentration.
- To compare typhoid cases across different countries and regions.
- To examine the influence of rainfall and water source types on disease occurrence.
- To evaluate the effect of healthcare access and GDP per capita on public health outcomes.
- To create an interactive Power BI dashboard for easy analysis and decision-making.
- To identify patterns that can support water management and disease prevention strategies.

4. Introduction

Water is one of the most essential natural resources for human survival, economic development, and environmental sustainability. Access to safe and clean drinking water is a fundamental requirement for maintaining public health and preventing the spread of infectious diseases. However, rapid industrialization, urbanization, agricultural activities, and improper waste disposal have significantly contributed to water pollution across many regions of the world. Contaminated water sources often contain harmful bacteria, chemicals, heavy metals, and other pollutants that pose serious risks to human health.

Waterborne diseases remain a major public health concern, particularly in developing countries where access to clean water and proper sanitation facilities is limited. Among these diseases, typhoid fever is one of the most common and dangerous illnesses caused by the consumption of contaminated food and water. The prevalence of typhoid is influenced by several factors, including water quality, sanitation practices, healthcare accessibility, environmental conditions, and socioeconomic status.

With the increasing availability of environmental and health-related data, data analytics and visualization tools have become valuable for understanding complex relationships between pollution and disease outbreaks. Analyzing such data can help governments, healthcare organizations, and policymakers make informed decisions regarding water management, public health interventions, and disease prevention strategies.

This project focuses on exploring the relationship between water pollution indicators and typhoid disease cases using the **Water Pollution and Disease** dataset obtained from Kaggle. The dataset includes information related to water quality parameters such as bacteria count, contaminant levels, dissolved oxygen, lead concentration, access to clean water, rainfall, healthcare access index, GDP per capita, water source types, water treatment methods, and reported typhoid cases across different countries and years.

To effectively analyze and interpret the data, an interactive dashboard was developed using Microsoft Power BI. The dashboard utilizes various visualization techniques, including KPI cards, donut charts, treemaps, line charts, column charts, and slicers, to present meaningful insights and trends. Through exploratory data analysis, the project aims to identify patterns, compare disease prevalence across regions, and understand how environmental and socioeconomic factors influence typhoid outbreaks.

The findings from this analysis can contribute to a better understanding of the impact of water pollution on public health and highlight the importance of improving water quality, sanitation systems, and healthcare accessibility to reduce the occurrence of waterborne diseases.

5. Workflow Structure

Step 1: Dataset Collection

- Downloaded the Water Pollution and Disease dataset from Kaggle.
- Verified data quality and structure.

Step 2: Data Import

- Imported dataset into Power BI Desktop.

Step 3: Data Cleaning

- Checked missing values.
- Corrected data types.
- Removed duplicate records.
- Formatted numerical columns.

Step 4: Data Transformation

- Created calculated fields and measures.
- Organized data for visualization.

Step 5: Dashboard Development

- Added KPI Cards.
- Added Donut Charts.
- Added Treemap Visualization.
- Added Line and Column Charts.
- Added Interactive Slicers.

Step 6: Analysis

- Examined relationships between water quality indicators and typhoid cases.
- Compared trends across countries and years.

Step 7: Reporting

- Documented findings and conclusions.

6. Methodology

6.1 Data Preparation

The dataset was loaded into Power BI and processed using Power Query Editor. Missing values and inconsistencies were handled to ensure accurate analysis.

6.2 Exploratory Data Analysis (EDA)

EDA was performed to understand:

- Distribution of water quality indicators.
- Disease occurrence patterns.
- Country-wise comparisons.
- Temporal trends.

6.3 Visualization Approach

The dashboard combines multiple visualizations to present information from different perspectives.

6.4 KPI Monitoring

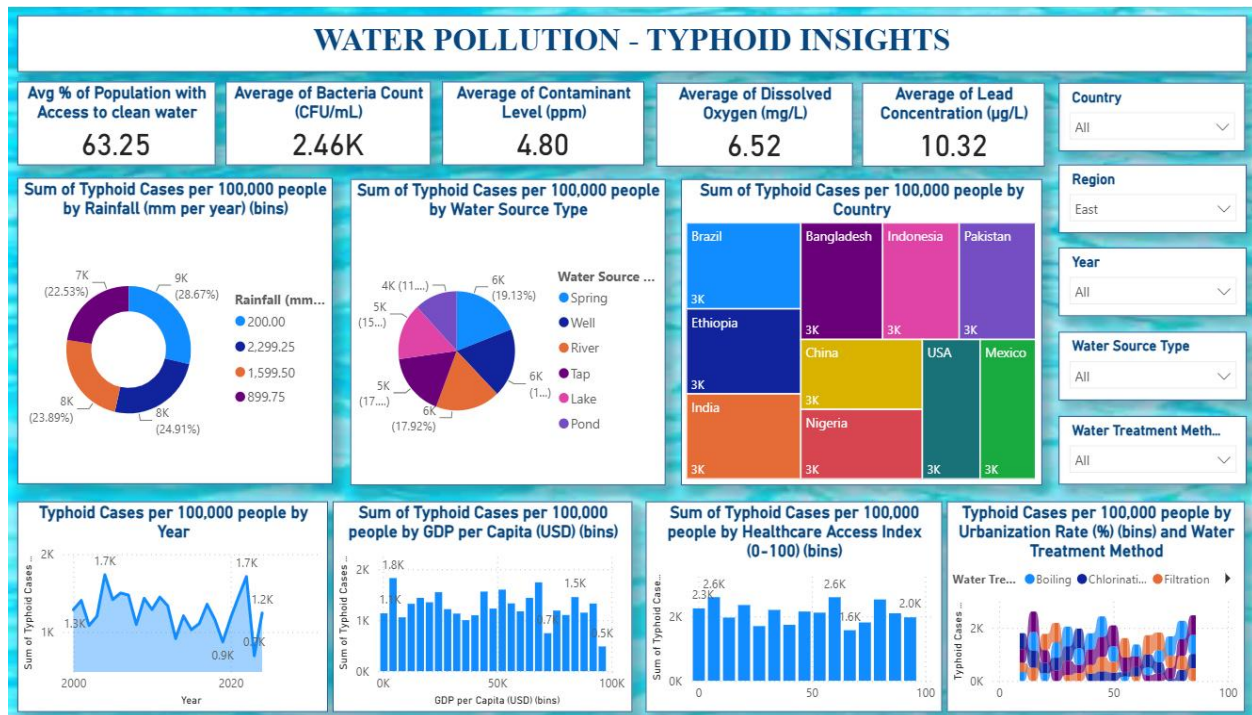
Key indicators were summarized using KPI cards for quick understanding.

6.5 Interactive Filtering

Slicers were used to filter results by:

- Country
- Region
- Year
- Water Source Type
- Water Treatment Method

7. Dashboard Description



1. KPI Cards

Avg % of Population with Access to clean water	Average of Bacteria Count (CFU/mL)	Average of Contaminant Level (ppm)	Average of Dissolved Oxygen (mg/L)	Average of Lead Concentration (µg/L)
64.61	2.49K	4.95	6.49	10.05

The top section of the dashboard presents key performance indicators that summarize important environmental metrics:

- Avg % of Population with Access to Clean Water (64.61%)

This indicator represents the average percentage of population having access to safe drinking water. Lower values may indicate higher vulnerability to waterborne diseases like typhoid.

- Average Bacteria Count (CFU/mL) – 2.49K

This shows the average microbial contamination level in water sources. Higher bacterial counts increase the risk of infectious diseases.

- Average Contaminant Level (ppm) – 4.95

This metric represents chemical contamination in water. Elevated contaminant levels may indirectly affect public health.

- Average Dissolved Oxygen (mg/L) – 6.49

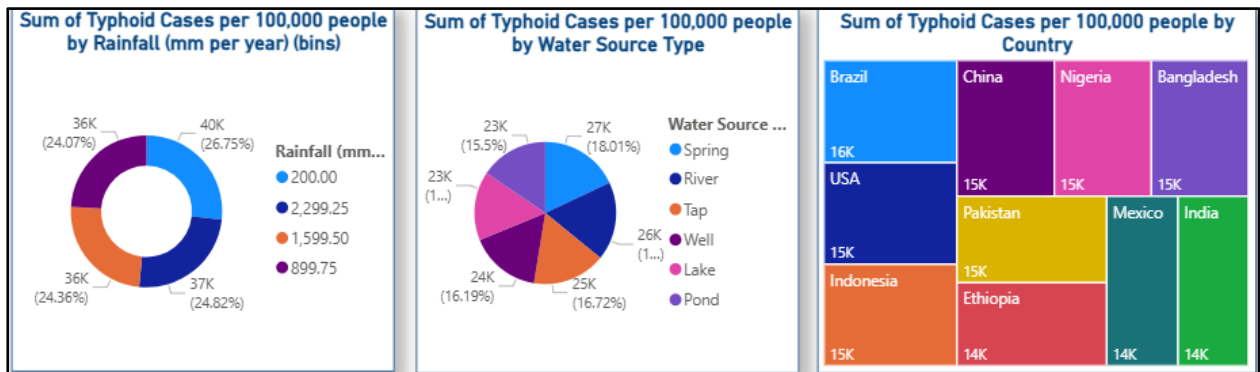
Dissolved oxygen indicates water quality. Lower oxygen levels can reflect polluted water conditions.

- Average Lead Concentration ($\mu\text{g/L}$) – 10.05

Lead is a harmful heavy metal. Higher concentrations indicate poor water treatment and environmental pollution.

These KPIs provide a quick overview of overall environmental health conditions.

2. Environmental Factor Analysis Visuals



- Donut Chart – Typhoid Cases by Rainfall (mm per year) (Bins)

This visualization groups rainfall into bins and shows the distribution of typhoid cases across rainfall levels.

Purpose:

- To analyze whether higher rainfall contributes to increased disease spread.
- Heavy rainfall may lead to water contamination and flooding, increasing infection risk.

- Pie Chart – Typhoid Cases by Water Source Type

Water sources include: Spring, River, Tap, Well, Lake, Pond

Purpose:

- To identify which type of water source is most associated with higher typhoid cases.
- Untreated sources like ponds and rivers may show higher disease contribution.

- Treemap – Typhoid Cases by Country

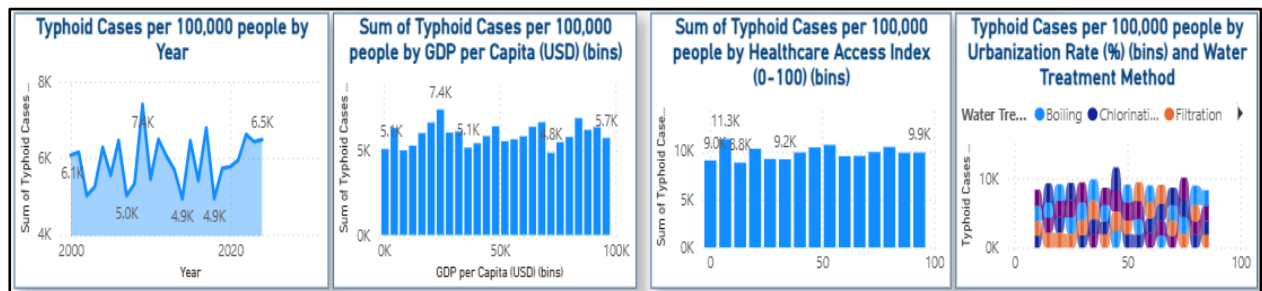
This treemap shows country-wise distribution of typhoid cases.

Purpose:

- To identify high-burden countries.
- Helps in geographical risk assessment.
- Larger blocks indicate higher total cases.

Countries like Brazil, China, Nigeria, Bangladesh, India, etc., are compared visually.

3. Temporal & Socio-Economic Analysis



- **Line Chart – Typhoid Cases per 100,000 People by Year**

This chart shows trends in typhoid cases over time.

Purpose:

- To detect increasing or decreasing patterns.
- To observe whether public health interventions have reduced cases.

- **Bar Chart – Typhoid Cases by GDP per Capita (Bins)**

GDP values are grouped into ranges and compared with typhoid incidence.

Purpose:

- To analyze economic influence on disease prevalence.
- Lower GDP countries may show higher disease burden.

- **Bar Chart – Typhoid Cases by Healthcare Access Index (0–100 Bins)**

Healthcare access index is grouped into ranges and analyzed against disease cases.

Purpose:

- To evaluate impact of healthcare availability.
- Better healthcare systems are expected to reduce disease rates.

- **Clustered Visualization – Urbanization Rate & Water Treatment Method vs Typhoid**

This visual analyzes: Urbanization rate bins, Water treatment methods (Boiling, Chlorination, Filtration)

Purpose:

- To examine how treatment methods influence disease across different urbanization levels.
- To identify the most effective water treatment method.

4. Interactive Filters (Slicers)



The right panel contains interactive filters that allow dynamic data exploration:

- **Country Filter**
Allows users to select specific country for analysis.
Updates all visuals accordingly.
- **Region Filter**
Filters data by geographical region (North, South, East, West, Central).
Enables regional comparison.
- **Year Filter**
Allows time-based analysis for specific years.
Useful for trend analysis.
- **Water Source Type Filter**
Filters data based on type of water source.
Helps analyze contamination impact.
- **Water Treatment Method Filter**
Allows comparison of treatment methods such as: Boiling, Chlorination, Filtration

Analytical Importance of Filters

The filters make the dashboard:

- Interactive
- Comparative
- Drill-down enabled
- Suitable for policy-level analysis

Users can:

- Select a country
- Analyze its GDP
- Compare healthcare access
- Evaluate rainfall impact
- Study treatment method effectiveness

This enhances decision-making capability.

The dashboard integrates environmental, economic, and healthcare indicators to provide a multidimensional analysis of typhoid incidence. It supports:

- Pattern identification
- Risk factor analysis
- Policy planning
- Public health monitoring

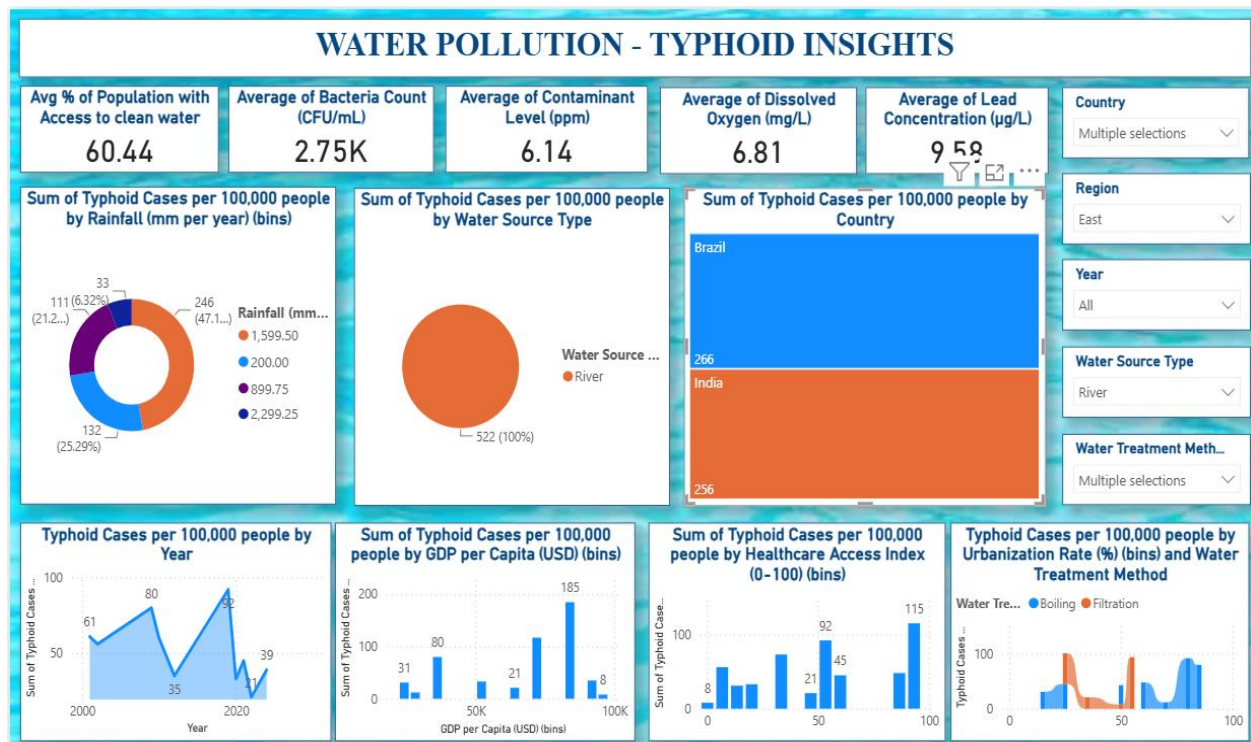
It serves as a decision-support tool for epidemiologists, government agencies, and environmental researchers.

Example: Comparative Country Analysis

To demonstrate the interactive analytical capability of the dashboard, a filtered analysis was performed by selecting two countries (Brazil and India) under the “Country” filter, along with the “River” water source type and the “East” region.

Observations:

1. The treemap visualization shows that Brazil and India contribute significantly to typhoid cases under river-based water sources.
2. When filtered to River as the primary water source, the total typhoid cases per 100,000 people show concentrated distribution, indicating that untreated river water may increase infection risk.
3. The rainfall-based donut chart suggests moderate to high rainfall bins contribute more cases in these selected countries.
4. The GDP bin chart indicates economic variability between the two countries, which may influence sanitation infrastructure and healthcare accessibility.
5. The healthcare access index bins show uneven distribution, suggesting healthcare availability differs across filtered regions.



Interpretation:

The comparative filtering reveals that dependence on river water sources in specific regions can be associated with higher typhoid incidence. Differences in healthcare access and economic status may further influence disease burden.

This demonstrates that the dashboard effectively supports drill-down analysis and cross-country comparison for policy evaluation.

8. Results

- Countries such as Brazil, China, Nigeria, Bangladesh, and the USA show relatively high typhoid case counts.
- Typhoid cases vary across different rainfall categories.
- Water source type influences disease occurrence.
- Healthcare access and economic conditions show noticeable relationships with disease prevalence.
- Disease trends fluctuate over time rather than remaining constant.

9. Conclusion

The dashboard successfully demonstrates the relationship between water pollution indicators and typhoid disease cases. Factors such as contaminated water sources, bacteria levels, rainfall patterns, healthcare accessibility, and economic conditions influence disease prevalence. The interactive Power BI dashboard provides a user-friendly platform for exploring these relationships and supports data-driven decision-making in environmental and public health management. Improved water quality, increased healthcare access, and effective treatment methods can contribute significantly to reducing disease outbreaks.

10.Reference

- Khushi Kyad. (2023). *Water Pollution and Disease* [Dataset]. Kaggle. <https://www.kaggle.com/datasets/khushikyad001/water-pollution-and-disease>
- Microsoft Learn. (2025). *Power BI documentation*. <https://learn.microsoft.com/en-us/power-bi/>

11.Report Summary

This project focused on analyzing the impact of water pollution on typhoid disease cases using Power BI. The dataset was collected from Kaggle and processed through data cleaning, transformation, and exploratory analysis. Interactive visualizations including KPI cards, donut charts, treemaps, line charts, and column charts were developed to study relationships among water quality indicators, rainfall, healthcare access, GDP, and disease occurrence. The dashboard revealed that water quality and socioeconomic factors significantly influence typhoid prevalence. The project demonstrates how business intelligence tools can be used to convert raw environmental data into meaningful insights that support public health planning and decision-making.